MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG1DSCSTA100 - Fundamentals of Statistics and Data Visualisation

Duration: 1Hour Maximum: 30 Marks

PRACTICAL EXAMINATION

From the following eight questions, students must answer any **five** questions. Each question carries **6** marks.

Solve the questions using spreadsheet in computer lab . The data in the question along with their answer must be written in the answer paper

1. A meteorologist wants to calculate the average temperature over a period to give a sense of the general weather conditions. For this, he recorded daily temperature in Fahrenheit for two weeks and obtained the data as 70, 76, 68, 71, 69, 76, 77, 70, 68, 65, 71, 72, 76 and 75. Help him to do this by calculating arithmetic mean and harmonic mean of the data [Apply][CO8]

2. Companies want to examine salary of the workers to understand overall compensation trends. If 15 workers earn in rupees 9000, 12000, 11500, 13500, 9500, 10000, 14500, 15000, 12800, 11400, 14700, 18000, 16500, 16400 and 12000 per month, find the median and mode for the data [Analyse][CO8]

3. A manager records the number of overtime hours worked by the employees in a week as 5, 7, 6.5, 3.5, 4, 2.5, 1.5, 2, 4, 5, 3, 7, 2, 4.5, 3.5. Find the geometric mean and harmonic mean of the data [Understand][CO7]

4. A quality control inspector counts the number of defects found in batches of products as 3, 7, 4, 3, 2, 7, 8, 4, 2, 4, 1, 6, 5, 7 ,4, 8, 3, 6, 5, 6, 9, 1, 6, 3, 6, 7, 6, 4. Find mode and arithmetic mean of the data.[Apply][CO8]

5. A sociologist wants to study income distribution at a place. He records income of different people as 15000, 25000, 35000, 17000, 32000, 33000, 25000, 20000, 30000, 28000, 32000, 31000, 29000, 26000, 27000, 19000, 23000. Find median and harmonic mean of the data. [Analyse][CO8]

6. A demographer wants to analyse the age distribution of people in a village. He collects data and recorded the ages of some people as 19, 24, 23, 43, 27, 28, 26, 34, 39, 20, 37, 38, 29, 38, 27, 40, 38, 25, 38, 20, 35, 37, 38. Find the mode and harmonic mean of the data [Understand][CO7]

7. The following data shows the yield of wheat in kilograms from 20 experimental plots: 38.5, 62.3, 46.3, 55.8, 52.3, 40.5, 38.6, 45.8, 50.7, 44.8, 43.8, 42.1, 54.9, 53.9, 41.8, 65.7, 34.9, 40.6, 47, 45. Find arithmetic mean and median for the data [Apply][CO8]

8. A textile company sells fashion outfits for men. Their net sales for different months in a year are 55, 67, 38, 29, 47,89,100, 56, 55, 35, 88, 89. Find median and harmonic mean of the data [Understand][CO7]

9. The number of tourists who visited India in a particular month from different countries are 150, 176, 230, 238, 178, 327, 339, 327, 267, 256, 352, 354, 327, 400, 386, 178, 197, 243, 134, 167, 186, 327, 139. Calculate mode and median for the data [Apply][CO8]

10. The following data shows the number of house wives doing part time jobs in different wards of a Panchayat as 12, 23, 35, 20, 28, 17, 18, 30, 27, 26, 33, 22, 29, 21, 29, 28, 27, 25, 20, 26. Find arithmetic mean and median for the data. [Apply][CO8]

11. A hospital records the waiting times in minutes for patients in the emergency room. Let the waiting times be 4, 3, 7, 2, 8, 10, 12, 3. Find geometric mean and harmonic mean of the data .[Analyse][CO8]

12. The following data gives the amount of money in rupees, spent on stationary items by some housewives in a village, during a week. 350, 400, 450, 340, 360, 310, 450, 550, 530, 560, 390, 520, 460, 440, 420. Find the range of the data . [Apply][CO8]

13. The purity of 20 gold coins were examined and the readings obtained in carat are as 20, 20.4, 19.6, 22.8, 23.4, 22.9, 23.9, 20.6, 20.4, 22.7, 20.1, 19.9, 21.7, 22.5, 21, 23.6, 22.8, 21.4, 20.8, 20.4. Find range and Quartile Deviation for the data.[Apply][CO8]

14. The time in hours spent by 20 students on their mobile phone on a day were recorded as 2, 3.6, 1.8, 3.5, 2.7, 2.5, 3, 4, 5.3, 3.9, 3.6, 4.2, 1.5, 2, 2.3, 2.8, 3, 3.4, 3.5, 3. Find mean deviation and quartile deviation of the data [Apply][CO8]

15. A hospital wants to know the variation in the data related to the ages of doctors and the ages were 36, 38, 29, 33, 32, 36, 30, 29,

31, 34, 36, 28, 30, 35, 37, 32, 35, 37. Help them to do it using SD and range.[Apply][CO8]

16. A company surveys employees about their daily commute times in minutes as 10, 14, 8, 16, 15, 17, 20, 22, 18, 16, 19, 16, 10, 20, 22, 32, 31, 26, 25, 24, 21, 27. Find quartile deviation and SD of the data.[Understand][CO8]

17. The family size of 16 students in a class are 3, 5, 2, 4, 5, 2, 2, 3, 4, 5, 3,3, 2, 4, 4, 5. Find mean deviation and SD.[Analyse][CO8]

18. The marks of students in a class, out of 25, were 24, 25, 14, 18, 17, 15, 22, 21, 20, 16, 15, 16, 18, 19, 22, 21, 20, 23. Find quartile deviation and mean deviation.[Apply][CO8]

19. A company tracks the working hours of employees in a day as 7, 6, 10, 9, 8, 7, 11, 12, 9, 7, 6, 5, 9, 10, 11, 12, 11. Find coefficient of variation for the data. [Understand][CO8]

20. The number of vehicles passing a certain junction on a busy day in 2 minutes interval are 10, 14, 16, 17, 18, 16, 14, 18, 12, 11, 14, 13, 15, 16, 15, 13, 18, 20. Find SD of the data.[Apply][CO8]

21. A teacher asked the students to complete 25 pages of a record book. 12 students have completed only 12, 15, 9, 16, 15, 14, 10, 19, 20, 17, 12, 17 pages. Find quartile deviation and SD of the data.[Analyse][CO8]

22. Attendance of students in a class during a month are 15, 20, 18, 17, 16, 22, 20, 19, 15, 19, 18, 20, 21, 20, 17, 16. Find skewness of the data.[Remember][CO8]

23. BMI values of some patients in a hospital were measured as 18.2, 19, 20.4, 21, 22.8, 19.9, 23.5, 24.7, 21.8, 23.3, 24.5, 21.5, 20.6, 20.5, 20.6. Find skewness of the data.[Analyse][CO8]

24. The number of patients admitted in different hospitals who are affected by TB are 5, 8, 11, 13, 9, 3, 2, 4, 10, 11, 16, 15, 18, 14, 11, 13, 8, 9, 10. Find skewness and kurtosis of the data.[Apply][CO8]

25. The number reels in instagram put forward by some students in a class in a week are 5, 6, 4, 10, 11, 9, 3, 9, 8, 10, 12, 6, 4, 8, 5, 11, 5, 9, 8, 6, 7. Find kurtosis of the data.[Analyse][CO8]

26. Calculate Karl Pearson correlation coefficient between ages (x) and marks (y) of students

age	10	12	13	14	16	17	18
mark	24	20	23	24	18	19	20

[Analyse][CO8]

27. Calculate Karl Pearson correlation coefficient between length (x) in cm and weight (y) in gms of 9 articles in a factory

length	7	12	10	20	16	9	10	9	12
weight	20	10	13	24	18	19	12	16	14

[Apply][CO8]

28. The monthly sales of items in a shop is as follows . Draw a simple bar diagram for this

month	Jan	Feb	Mar	Apr	May	June
sales	1500	2000	2500	1200	3000	1000

[Apply][CO8]

29. The following data shows the amount of fruits (in kg) consumed by different families in a village. Draw a simple bar diagram for this. [Analyse][CO8]

fruit	Apple	Banana	Grape	Orange	lemon	peach	papaya
freq	30	25	50	60	70	40	30

30. The following data shows the market shares of different smart phone brands. Draw a pie diagram for this

brand	A	B	C	D	E	F	G
share	30	45	70	100	25	38	60

[Apply][CO8]

31. The following data gives the distribution of expenses in a family. Draw a pie diagram for this

category	housing	food	travel	cleaning	education	hobby
expense	20000	10000	25000	5000	7000	10000

[Analyse][CO8]

32. The following data shows the marks of students in English in a class. Draw a frequency curve for this

marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
freq	7	12	15	20	5	3	2

[Apply][CO8]

33. The following data shows the waiting times (in minutes) of some students at a bus junction. Draw a frequency polygon for this data

time	0-3	3-6	6-9	9-12	12-15	15-18	18-21	21-24
freq	5	11	15	16	9	7	4	2

[Apply][CO8]

34. The daily internet usage time (in minutes) of some college students are 20, 17, 25, 35, 32, 20, 40, 55, 60, 44, 43. Draw a box plot for this data.[Analyse][CO8]

35. The data on journey time of 80 students to reach school from their place of residence is given. Draw ogives for the data

time	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
No. of students	5	12	16	20	14	9	2	2

[Apply][CO8]

36. The following data shows the holding of shares by different persons in an IT company. Draw a less than ogive for the data

shares	0-4	4-8	8-12	12-16	16-20	20-24	24-28	28-32
freq	10	22	30	40	25	17	8	7

[Apply][CO8]

37. The growth in mm of certain plants during 2 months are as follows. Draw a greater than ogive for this.

growth	2-6	6-10	10-14	14-18	18-22	22-26	26-30	30-34
freq	14	26	38	45	33	22	16	7

[Apply][CO8]

(5 x 6 = 30)